

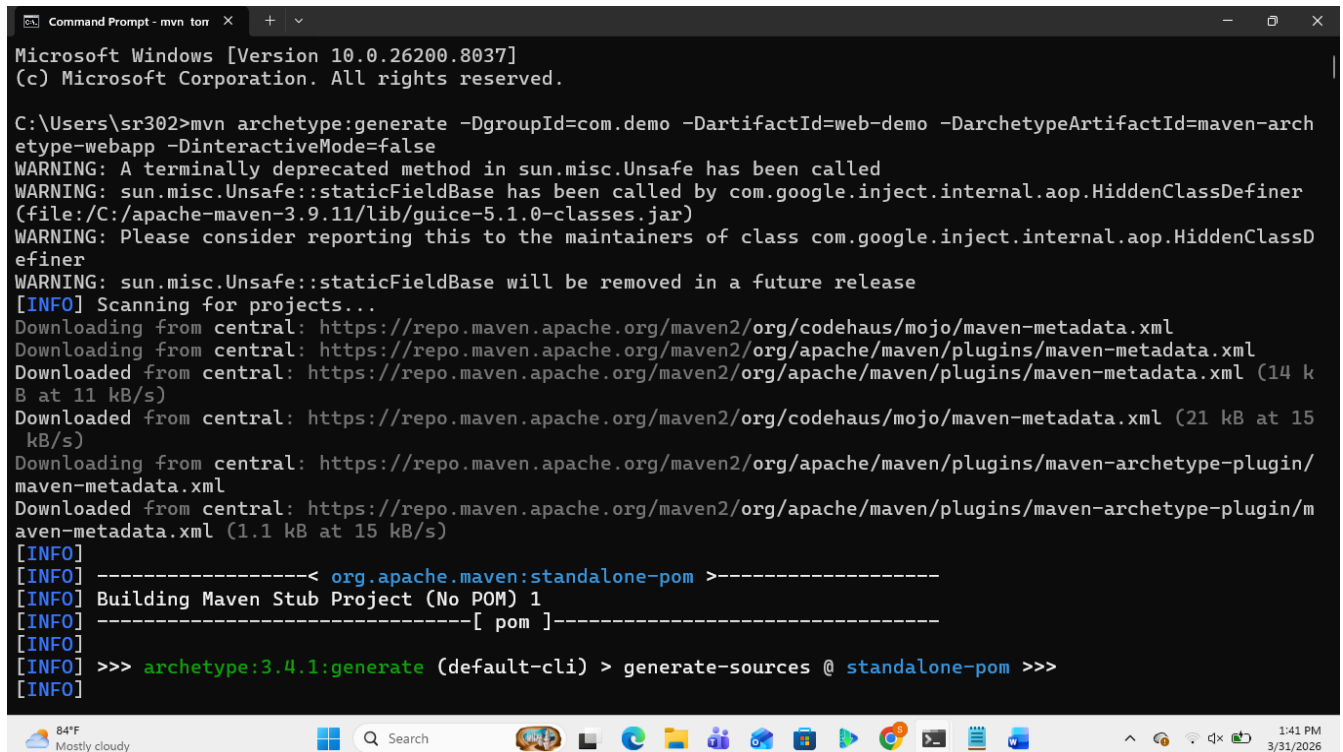
# INT-332 Practical-3

## Maven Practical

### Exercise 1: Maven WAR File .

#### Step 1: Create Maven Project

`mvn archetype:generate -DgroupId=com.demo -DartifactId=web-demo -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false`



```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sr302>mvn archetype:generate -DgroupId=com.demo -DartifactId=web-demo -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner
(file:/C:/apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml (14 kB at 11 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml (14 kB at 11 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml (21 kB at 15 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-metadata.xml (1.1 kB at 15 kB/s)
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
```

```
Command Prompt - mvn tor X + v
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 12 MB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.0/maven-archetype-webapp-1.0.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.0/maven-archetype-webapp-1.0.jar (3.9 kB at 19 kB/s)
[INFO] -----
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] -----
[INFO] Parameter: basedir, Value: C:\Users\sr302
[INFO] Parameter: package, Value: com.demo
[INFO] Parameter: groupId, Value: com.demo
[INFO] Parameter: artifactId, Value: web-demo
[INFO] Parameter: packageName, Value: com.demo
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: C:\Users\sr302\web-demo
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 7.771 s
[INFO] Finished at: 2026-03-31T12:22:07+05:30
[INFO] -----
```

## Step 2: Package Application:-

Mvn clean package

```
Command Prompt - mvn tor X + v
C:\Users\sr302\web-demo>mvn clean install
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner
(file:/C:/apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.demo:web-demo >-----
[INFO] Building web-demo Maven Webapp 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.4.0/maven-war-plugin-3.4.0.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.4.0/maven-war-plugin-3.4.0.pom (8.4 kB at 1.4 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.4.0/maven-war-plugin-3.4.0.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.4.0/maven-war-plugin-3.4.0.jar (83 kB at 153 kB/s)
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ web-demo ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ web-demo ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource from src/main/resources to target/classes
```

```
Command Prompt - mvn tom - X + v
[INFO] --- install:3.1.2:install (default-install) @ web-demo ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18.pom (2.9 kB at 35 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver/1.9.18/maven-resolver-1.9.18.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver/1.9.18/maven-resolver-1.9.18.pom (22 kB at 314 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.18/maven-resolver-api-1.9.18.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.18/maven-resolver-api-1.9.18.pom (2.7 kB at 39 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18.jar (196 kB at 488 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.18/maven-resolver-api-1.9.18.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.18/maven-resolver-api-1.9.18.jar (157 kB at 480 kB/s)
[INFO] Installing C:\Users\sr302\web-demo\pom.xml to C:\Users\sr302\.m2\repository\com\demo\web-demo\1.0-SNAPSHOT\web-demo-1.0-SNAPSHOT.pom
[INFO] Installing C:\Users\sr302\web-demo\target\web-demo-1.0-SNAPSHOT.war to C:\Users\sr302\.m2\repository\com\demo\web-demo\1.0-SNAPSHOT\web-demo-1.0-SNAPSHOT.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 35.910 s
```

### Step 3: Modify index.jsp Program:-

With path : src/main/java/com/example/App.java

```
index.jsp X pom.xml index.jsp
File Edit View

<html>
<body>
<h1>Hello! This webpage is running using Maven.</h1>
</body>
</html>
```

## Step 4: Add dockerfile-maven-plugin in pom.xml:-

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
```

```
<modelVersion>4.0.0</modelVersion>
```

```
<groupId>com.demo</groupId>
```

```
<artifactId>web-demo</artifactId>
```

```
<version>1.0-SNAPSHOT</version>
```

```
<packaging>war</packaging>
```

```
<name>web-demo Maven Webapp</name>
```

```
<dependencies>
```

```
<dependency>
```

```
<groupId>junit</groupId>
```

```
<artifactId>junit</artifactId>
```

```
<version>3.8.1</version>
```

```
<scope>test</scope>
```

```
</dependency>
```

```
</dependencies>
```

```
<build>
```

```
<plugins>
```

```
<plugin>
```

```
<groupId>org.apache.tomcat.maven</groupId>
```

```
<artifactId>tomcat7-maven-plugin</artifactId>
```

```
<version>2.2</version>
```

```
<configuration>

<port>8080</port>

<path>/</path>

</configuration>

</plugin>

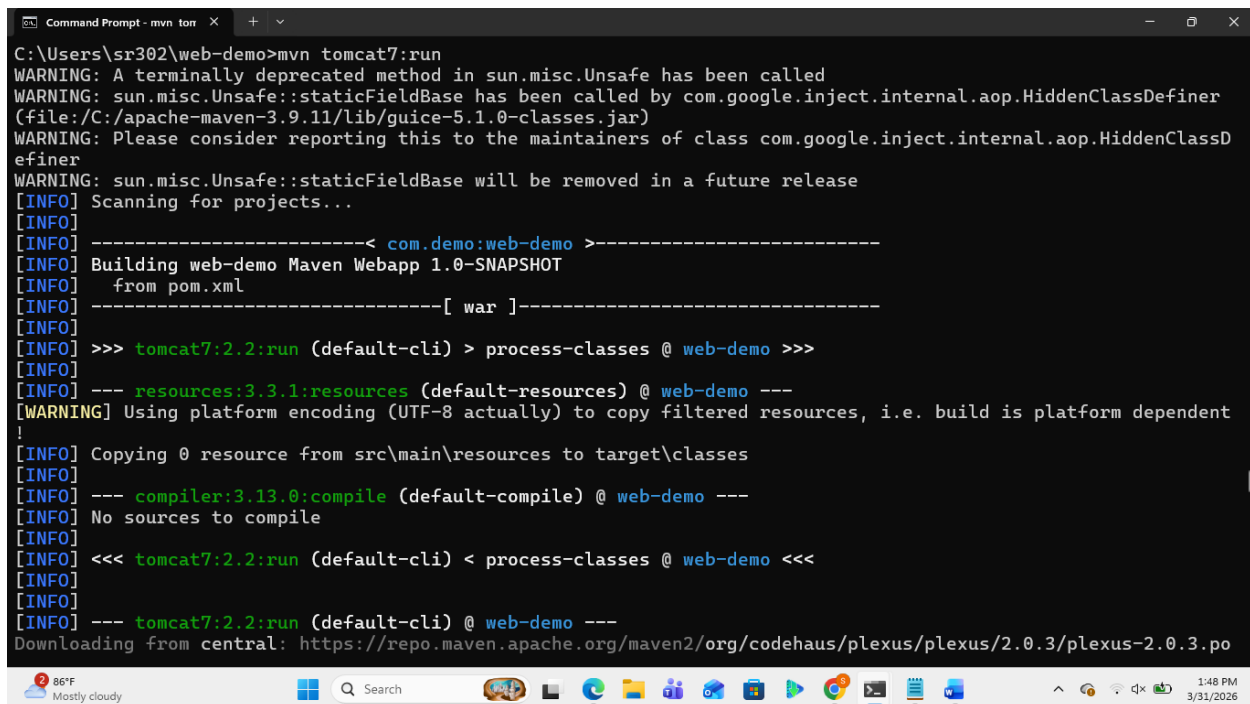
</plugins>

</build>

</project>
```

## Step 5: Run Docker Container:-

mvn tomcat7:run



```
Command Prompt - mvn tom X + v
C:\Users\sr302\web-demo>mvn tomcat7:run
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner
(file:/C:/apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassD
efiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.demo:web-demo >-----
[INFO] Building web-demo Maven Webapp 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO]
[INFO] >>> tomcat7:2.2:run (default-cli) > process-classes @ web-demo >>>
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ web-demo ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent
!
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ web-demo ---
[INFO] No sources to compile
[INFO]
[INFO] <<< tomcat7:2.2:run (default-cli) < process-classes @ web-demo <<<
[INFO]
[INFO] --- tomcat7:2.2:run (default-cli) @ web-demo ---
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/2.0.3/plexus-2.0.3.po
```

```
Command Prompt - mvn tomcat X + v
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-plugin-descriptor/2.2.1/maven-plugin-descriptor-2.2.1.jar (39 kB at 419 B/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/slf4j/jcl-over-slf4j/1.7.5/jcl-over-slf4j-1.7.5.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-monitor/2.2.1/maven-monitor-2.2.1.jar (10 kB at 111 B/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/slf4j/slf4j-api/1.7.5/slf4j-api-1.7.5.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-settings/2.2.1/maven-settings-2.2.1.jar (49 kB at 521 B/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/slf4j/jcl-over-slf4j/1.7.5/jcl-over-slf4j-1.7.5.jar (17 kB at 175 B/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/plexus/plexus-build-api/0.0.4/plexus-build-api-0.0.4.jar (6.8 kB at 72 B/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/slf4j/slf4j-api/1.7.5/slf4j-api-1.7.5.jar (26 kB at 275 B/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/jdt/core/compiler/ecj/4.2.2/ecj-4.2.2.jar (1.8 MB at 16 kB/s)
[INFO] Running war on http://localhost:8080/
[INFO] Creating Tomcat server configuration at C:\Users\sr302\web-demo\target\tomcat
[INFO] create webapp with contextPath:
Mar 31, 2026 12:34:53 PM org.apache.coyote.AbstractProtocol init
INFO: Initializing ProtocolHandler ["http-bio-8080"]
Mar 31, 2026 12:34:53 PM org.apache.catalina.core.StandardService startInternal
INFO: Starting service Tomcat
Mar 31, 2026 12:34:53 PM org.apache.catalina.core.StandardEngine startInternal
INFO: Starting Servlet Engine: Apache Tomcat/7.0.47
Mar 31, 2026 12:34:56 PM org.apache.coyote.AbstractProtocol start
INFO: Starting ProtocolHandler ["http-bio-8080"]
```